
LOW RANK ADAPTIVE FRICTION FOR MEMORY-EFFICIENT TRANSFORMER PRETRAINING

PREPRINT

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ABSTRACT

iKFAD is a recently proposed optimizer that closes the Adam–SGD performance gap observed in transformer model pre-training by replacing adaptive learning rates with adaptive friction in the momentum dynamics. Its limitation is that the full friction tensor $\xi \in \mathbb{R}^{m \times n}$ carries the same $\mathcal{O}(nm)$ memory overhead per layer as Adam’s second-moment buffer. We apply the rank-1 compression idea that Adafactor applies to Adam’s second moment, replacing iKFAD’s friction tensor ξ with a rank-1 outer-product factorization built from row and column momentum statistics, resulting in Rank-1 iKFAD (R-iKFAD). This reduces friction memory footprint from $\mathcal{O}(nm)$ to $\mathcal{O}(n + m)$ per layer which approximately halves iKFAD’s total optimizer state. We find that despite this massive decrease in signal, R-iKFAD maintains parity in performance with iKFAD, and consequently Adam on transformer pre-training tasks. Experiments on GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches iKFAD/Adam while nearly halving the memory footprint and being comparably robust to hyperparameters. We prove exponential convergence of the continuous-time dynamics and its splitting discretization under strong convexity.

1 Introduction

Adam [7] remains the workhorse optimizer in modern deep learning, combining gradient momentum with coordinate-wise second-moment scaling to accelerate convergence on complex loss landscapes. For a model with N scalar parameters, Adam tracks two auxiliary states (the first and second moment estimates), requiring a total optimizer memory footprint of $3N$ scalars (including parameters). At scale, this $3\times$ memory overhead presents a severe hardware bottleneck during large-model training. To mitigate this overhead, Adafactor [11] introduces a rank-1 factorisation of the second-moment matrix. For a matrix weight $W \in \mathbb{R}^{m \times n}$, Adafactor replaces Adam’s full second-moment matrix $V \in \mathbb{R}^{m \times n}$ with a factorized approximation:

$$\hat{V}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^\top R},$$

where $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$ track row-wise and column-wise squared gradients, respectively. By discarding the first-moment buffer, Adafactor reduces per-layer auxiliary state complexity from $\mathcal{O}(mn)$ to $\mathcal{O}(m + n)$ (achieving memory savings of up to 99.8% for large weight matrices) while maintaining competitive transformer pretraining performance.

An insightful paradigm for analyzing optimizer dynamics is continuous-time ordinary differential equations (ODEs) [1, 3], where updates emerge as discrete time-step approximations of continuous dynamical systems. For a parameter state

x , momentum p , and second-moment variable ζ , continuous-time Adam is expressed as:

$$\dot{x} = \frac{p}{\sqrt{\zeta} + \epsilon}, \quad (1)$$

$$\dot{p} = -\nabla f(x) - \gamma p, \quad (2)$$

$$\dot{\zeta} = [\nabla f(x)]^2 - \alpha \zeta. \quad (3)$$

While Adam incorporates parameter-wise adaptation into the *position* equation ($\dot{x}$) via $1/\sqrt{\zeta}$, Karoni et al. [5] demonstrated that coordinate-wise adaptation can alternatively be integrated into the *momentum* equation ($\dot{p}$). This is achieved by replacing the constant damping factor γ with a dynamic, coordinate-wise friction state ξ :

$$\dot{x} = p, \quad (4)$$

$$\dot{p} = -\nabla f(x) - (\gamma + \xi) \odot p, \quad (5)$$

$$\dot{\xi} = \frac{[p]^2}{\rho} - \alpha \xi. \quad (6)$$

These dynamics, termed *Individual Kinetic Friction Adaptive Descent* (iKFAD), requires storing a full adaptive friction tensor ξ alongside momentum p . It incurs an auxiliary footprint of $2N$ scalars, identical to Adam’s auxiliary storage. Originally introduced for scalar/global friction [4], iKFAD extends friction-based adaptation to individual parameters, matching Adam’s performance on standard vision and NLP benchmarks.

We formulate Adafactor as a continuous-time dynamical system similar to Adam. For a matrix weight $X \in \mathbb{R}^{m \times n}$, the dynamics are given by:

$$\dot{X} = -\frac{\nabla f(X)}{\sqrt{\frac{RC^\top}{\mathbf{1}_m^\top R} + \epsilon}}, \quad (7)$$

$$\dot{R} = [\nabla f(X)]^2 \mathbf{1}_n - \alpha R, \quad (8)$$

$$\dot{C} = \mathbf{1}_m^\top [\nabla f(X)]^2 - \alpha C, \quad (9)$$

where the explicit momentum is omitted in line with standard Adafactor conventions, and the rank-1 factorisation emerges as a steady-state approximation $\zeta \approx RC^\top / (\mathbf{1}_m^\top R)$.

The structural relationship between Adafactor and Adam directly mirrors that of our proposed method, *R-iKFAD*, to iKFAD. Just as Adafactor replaces Adam’s $\mathcal{O}(mn)$ second-moment state with low-rank vectors, we replace iKFAD’s full $\mathcal{O}(mn)$ friction tensor $\xi \in \mathbb{R}^{m \times n}$ with a rank-1 outer-product estimate:

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^\top R},$$

where $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$ aggregate row- and column-wise squared-momentum $[p]^2$ statistics. The resulting algorithm, **Rank-1 iKFAD (R-iKFAD)**, retains momentum and kinetic friction dynamics while compressing friction storage to $\mathcal{O}(m+n)$ per layer. This effectively halves iKFAD’s auxiliary memory overhead (Section D).

We emphasize two key distinctions:

1. **Performance target:** R-iKFAD is designed to maintain parity with iKFAD (which already matches Adam [5]), rather than strictly outperform it.
2. **Memory regime:** R-iKFAD maintains a middle ground between full Adam ($2N$ auxiliary states) and Adafactor (≈ 0 auxiliary states) by retaining full momentum dynamics at only $\approx N$ auxiliary states. This is a deliberate design choice: Adafactor is known to suffer from training instabilities in certain transformer regimes due to discarding momentum entirely [11].

2 Methodology: Rank-1 iKFAD

For a matrix parameter $X \in \mathbb{R}^{m \times n}$ with momentum $P \in \mathbb{R}^{m \times n}$, the continuous-time dynamics of R-iKFAD are given by:

$$\dot{X} = P, \quad (10)$$

$$\dot{P} = -\nabla f(X) - \left(\frac{RC^\top}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} \odot P \right) - \gamma P, \quad (11)$$

$$\dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (12)$$

$$\dot{C} = \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C, \quad (13)$$

where $\odot$ denotes the element-wise Hadamard product, and $[P]^2$ denotes element-wise squaring ($[P]_{ij}^2 = P_{ij}^2$). The scalar $\epsilon_{\text{stab}} > 0$ ensures numerical stability in the outer-product denominator.

The core motivation of R-iKFAD is to eliminate the prohibitive $\mathcal{O}(mn)$ state footprint of iKFAD’s full friction matrix $\xi \in \mathbb{R}^{m \times n}$ while retaining both full parameter momentum P and per-coordinate damping adaptation. By approximating the dense friction tensor as a rank-1 factorization $\xi = RC^\top / (\mathbf{1}_m^\top R + \epsilon_{\text{stab}})$, the state per layer scales as $\mathcal{O}(m + n)$ for friction vectors $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$, which track row- and column-wise kinetic energy accumulated by squared momentum terms $[P]^2$.

Splitting Discretization. To construct a stable discrete-time algorithm, we follow best practices for geometric numerical integration of Hamiltonian systems [8]. Following [5], we decompose the vector field into sub-flows that admit exact analytical integration. This operator splitting scheme preserves sub-step stability and geometric properties without requiring conservative step sizes h . The detailed sub-flow decomposition and closed-form update equations are provided in Appendix A.

Memory Complexity. For a network parameter matrix at layer ℓ given by $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$, let $N = \sum_\ell m_\ell n_\ell$ denote the total number of scalar model parameters. Table 1 breaks down the auxiliary optimizer memory footprint required on top of the model weights. For non-2D parameters (embeddings, biases, LayerNorm scales) that are not rank-1 factorizable, R-iKFAD falls back to element-wise iKFAD, contributing a negligible additional scalar per parameter. R-iKFAD requires approximately N auxiliary scalars in total, effectively halving the optimizer auxiliary memory footprint relative to both iKFAD and Adam.

Table 1: Comparison of auxiliary optimizer memory complexity (excluding parameters).

Optimizer	Auxiliary State Variables	Total Auxiliary Footprint
Adam	First moment M , Second moment V	$2N$ scalars
iKFAD	Momentum P , Friction tensor ξ	$2N$ scalars
R-iKFAD (ours)	Momentum P , Factors $(R_\ell, C_\ell)_\ell$	$N + \sum_\ell (m_\ell + n_\ell)$ scalars

Convergence Analysis. Although neural network loss landscapes are highly non-convex in practice, it is useful and standard to analyze the dynamics under strong convexity and standard smoothness conditions. We prove that under these conditions continuous-time R-iKFAD dynamics converges exponentially to the global minimum and for a sufficiently small step size h , a splitting discretization preserves this rate too. See Appendix E for Theorems and corresponding proofs.

3 Experiments

We set $\gamma = 0$ for both iKFAD and R-iKFAD throughout (denoted **iKFAD0** and **R-iKFAD0** respectively), following [5]; Section 3.2 confirms $\gamma > 0$ yields no benefit. We compare both against Adam and Adafactor across four benchmarks spanning image classification and language modelling at scales from 0.85M to 125M parameters. All hyperparameters are tuned via Optuna (TPE sampler) with equal sweep budgets of approximately 80 trials per optimizer per model; NanoGPT and TinyViT received substantially more trials. See Appendix B for full configurations. Just like iKFAD, the optimal scale of μ varies by several orders of magnitude between different tasks due to differing momentum scales. Therefore, we recommend a wide-range initial sweep and note this as the primary limitation of our method.

- **NanoGPT (0.85M)** [6] on Shakespeare character-level language modelling.
- **TinyViT (5M)** [12] on CIFAR-10 image classification.
- **DistilBERT (80M)** [2] on SST-2 sentiment classification (pretraining from scratch).
- **GPT2-S (125M)** [10] on OpenWebText pretraining (modded-nanoGPT architecture).

3.1 Main Results

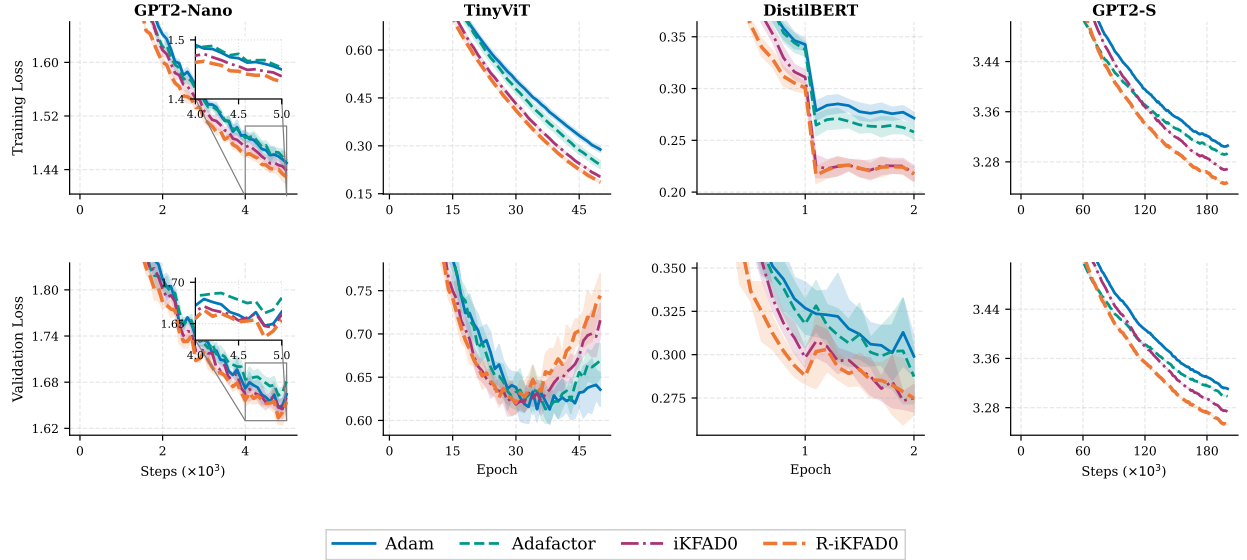


Figure 1: Training loss curves for all five benchmarks, averaged over 10 random seeds. R-iKFAD0 matches or improves upon iKFAD0 on all tasks, while using approximately half the optimizer memory. Both methods are competitive with Adam and Adafactor, which require comparable or greater memory. Shaded regions denote one standard deviation.

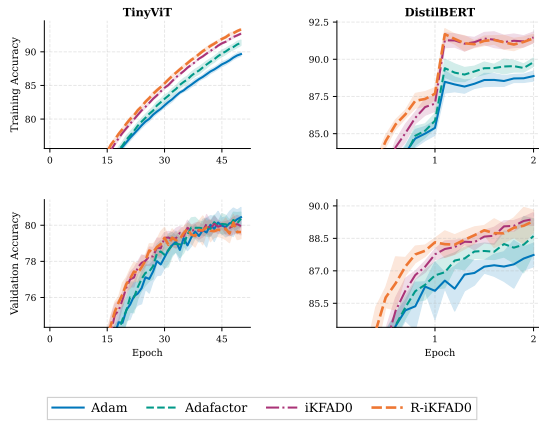
Figures 1 and 2 show training loss and accuracy across all benchmarks. R-iKFAD0 matches iKFAD0 in convergence speed and final performance while using approximately half the optimizer memory, confirming that the rank-1 friction approximation incurs no performance cost. Crucially, R-iKFAD0 is competitive with Adam throughout, consistent with the finding in [5] that adaptive friction achieves parity with adaptive learning rates on LLM pretraining. We do not claim R-iKFAD surpasses Adam; the goal is to preserve iKFAD’s performance parity with Adam while reducing its memory overhead. Table 2 summarises the lowest validation loss achieved during training for all methods. GPT2-S results use a single seed due to computational constraints; all other benchmarks report mean $\pm$ standard deviation over 10 seeds. The optimizer auxiliary state is the memory an optimizer adds on top of the model parameters. Table 5 in Appendix D reports the absolute values on all four benchmarks in MB (float32).

3.2 Ablation: $\gamma = 0$ versus $\gamma > 0$

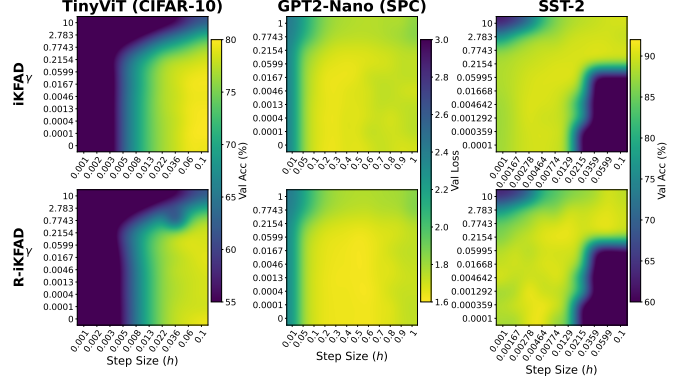
Figure 3 (Appendix C) shows that $\gamma > 0$ yields no consistent improvement over $\gamma = 0$ on any of the three benchmarks tested (NanoGPT, TinyViT, DistilBERT). This is consistent with the finding in [5] that the adaptive friction mechanism alone provides sufficient damping. We therefore recommend **R-iKFAD0** ($\gamma = 0$) as the practical default: it has one fewer hyperparameter.

3.3 Robustness to Hyperparameters

Figure 2 presents grid sweeps over step size h and friction coefficient γ for both R-iKFAD and iKFAD0. The high-performance regions in the γ - h plane are comparable for both methods on all three benchmarks. This robustness is practically important: it means that a coarse initial hyperparameter search is sufficient to identify a near-optimal configuration and that the method does not require fine-grained tuning of h or γ to perform well. The breadth of the high-performance plateau is a direct consequence of the adaptive kinetic friction mechanism. When momentum P increases due to a large step size h , rank-1 friction factors R and C automatically accumulate larger squared-momentum statistics, causing $\tilde{\xi}$ to increase in size and increase damping. This self-stabilizing feedback loop prevents step-size



(a) Accuracy averaged over 10 seeds. R-iKFAD0 matches iKFAD0 and Adam. Error bars denote one std.



(b) Two-dimensional γ - h sensitivity for R-iKFAD (top) and iKFAD (bottom). Each cell shows best performance at a fixed (h, γ) grid point; brighter = better. R-iKFAD retains comparably broad high-performance plateaus.

Figure 2: Accuracy/perplexity results and γ - h robustness sweeps for R-iKFAD0 and baselines.

overshooting across a wide range of h values. Consequently, R-iKFAD inherits the same plateau structure observed for iKFAD [5], confirming that rank-1 compression does not degrade the optimizer’s tolerance to aggressive hyperparameter choices.

Table 2: Lowest validation loss (mean $\pm$ std over 10 seeds; GPT2-S: single seed due to computational limitations). Lower is better. R-iKFAD0 matches or improves upon iKFAD0 across all benchmarks while using $\approx 2\times$ less optimizer memory on transformer models.

Dataset	Adam	Adafactor	iKFAD0	R-iKFAD0 (ours)
GPT2-Nano	1.643 ± 0.008	1.661 ± 0.007	1.640 ± 0.008	1.633 ± 0.011
TinyViT	0.593 ± 0.015	0.596 ± 0.010	0.606 ± 0.011	0.606 ± 0.012
DistilBERT	0.292 ± 0.005	0.280 ± 0.007	0.269 ± 0.005	0.268 ± 0.003
GPT2-S	3.301	3.287	3.263	3.242

4 Conclusion

We introduced **R-iKFAD**, an optimizer that applies rank-1 compression to iKFAD’s friction tensor to reduce per-layer friction storage from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$. By maintaining a full momentum buffer alongside low-rank friction vectors, R-iKFAD occupies an intermediate regime of “Adafactor with momentum” derived from provably convergent friction dynamics. The goal is not to surpass Adam, but to preserve iKFAD’s established performance parity with Adam while cutting auxiliary optimizer memory in half. Our experiments across GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches iKFAD in convergence speed, final accuracy, and hyperparameter robustness at half the auxiliary memory cost. On the theoretical side, we prove exponential convergence for both the continuous dynamics and its splitting discretization under strong convexity. Future work includes extending convergence bounds to the stochastic gradient case and validating performance on larger scale LLMs.

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A Update Rules for Splitting Discretization

To derive a practical discrete-time algorithm from the continuous-time dynamics (10)–(13), we use an operator-splitting approach [8]. The full vector field is decomposed into four sub-operators whose individual flows can be integrated analytically:

$$\begin{pmatrix} \dot{X} \\ \dot{P} \\ \dot{R} \\ \dot{C} \end{pmatrix} = \underbrace{\begin{pmatrix} P \\ 0 \\ 0 \\ 0 \end{pmatrix}}_A + \underbrace{\begin{pmatrix} 0 \\ -\nabla f(X) \\ 0 \\ 0 \end{pmatrix}}_B + \underbrace{\begin{pmatrix} -\frac{RC^\top}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} \odot P \\ -\frac{1}{\mu}[P]^2 \mathbf{1}_n - \alpha R \\ -\frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C \end{pmatrix}}_C + \underbrace{\begin{pmatrix} 0 \\ -\gamma P \\ 0 \\ 0 \end{pmatrix}}_D. \quad (14)$$

Sub-flows A, B, and D integrate analytically: A gives position drift; B applies the gradient kick; D applies linear friction (becoming the identity when $\gamma = 0$, i.e., R-iKFAD0). Sub-flow C couples momentum P and friction factors (R, C) and is integrated semi-analytically via an internal symmetric freeze-then-update scheme. The composition applied in practice is a first-order Lie–Trotter splitting:

$$\Phi_h = \Phi_h^D \circ \Phi_h^C \circ \Phi_h^A \circ \Phi_h^B,$$

where sub-steps are executed in the sequence $B \rightarrow A \rightarrow C \rightarrow D$ (BACD) [5]. Position updates use the gradient-kicked momentum $P^{(B)}$, yielding a symplectic-Euler structure for the (X, P) subsystem that prevents secular drift. Table 3 summarizes the explicit analytical update rules for each sub-step across the complete BACD sequence over time step h .

Within sub-flow C, no closed-form joint solution exists for coupled P and (R, C) . We apply a symmetric half-damp sequence: evaluating $\exp(-\frac{h}{2}\tilde{\xi})$ twice around an exact exponential ODE factor update improves integration accuracy to $\mathcal{O}(h^2)$ within sub-flow C alone. Using the post-first-half-damp momentum $P^{(1)}$ for factor updates reduces forcing magnitude and prevents over-accumulation.

Table 3: Analytical sub-step update rules for the BACD splitting sequence over time step h .

Sub-step	Update Equation
B	$P^{(B)} = P_n - h \nabla f(X_n)$
A	$X_{n+1} = X_n + h P^{(B)}$
C	$\tilde{\xi}_n = \frac{R_n C_n^\top}{\mathbf{1}_m^\top R_n + \epsilon_{\text{stab}}}$ $P^{(1)} = P^{(B)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_n\right)$ $R_{n+1} = e^{-\alpha h} R_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} [P^{(1)}]^2 \mathbf{1}_n$ $C_{n+1} = e^{-\alpha h} C_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} \mathbf{1}_m^\top [P^{(1)}]^2$ $\tilde{\xi}_{n+1} = \frac{R_{n+1} C_{n+1}^\top}{\mathbf{1}_m^\top R_{n+1} + \epsilon_{\text{stab}}}$ $P^{(2)} = P^{(1)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_{n+1}\right)$
D	$P_{n+1} = P^{(2)} \odot e^{-\gamma h} \quad (\text{identity when } \gamma = 0)$

B Optimal Hyperparameters

Across models and tasks, the best-performing scales for μ (iKFAD0, R-iKFAD0) varied by several orders of magnitude. This variation is likely related to the scale of the momentum variables during training, since μ controls the strength of the adaptive friction terms. Normalizing the momentum variables across iterations could potentially reduce this sensitivity, but we leave this for future work.

Experimental configurations:

- **ResNet-18 – CIFAR-10:** batch size 64, 40 epochs, 100 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **TinyViT – CIFAR-10:** batch size 128, 25 epochs, 100 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **DistilBERT – SST-2:** batch size 16, 2 epochs, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-Nano – Shakespeare:** batch size 16, 2001 steps, 500 trials. $h \in [10^{-6}, 5 \times 10^{-1}]$, $\alpha \in [10^{-5}, 10]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-S – OpenWebText:** batch size 16, 50001 steps, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and $\mu \in [10^{-9}, 10^{-2}]$ for R-iKFAD0.

Table 4: Optimized hyperparameters for all experiments. All iKFAD0 and R-iKFAD0 entries have $\gamma = 0$. Dashes indicate parameters not used by the method.

	h	α	μ / ρ	β_1	β_2
ResNet-18 (CIFAR-10)					
iKFAD0	0.0462	0.1036	7.2425×10^{-7}	–	–
R-iKFAD0	0.0566	0.0602	4.6430×10^{-6}	–	–
Adam	8.0000×10^{-4}	–	–	0.9700	0.9900
Adafactor	2.5447×10^{-4}	–	–	0.8984	–
TinyViT (CIFAR-10)					
iKFAD0	0.0725	0.0899	1.2643×10^{-5}	–	–
R-iKFAD0	0.0860	0.5327	9.4765×10^{-7}	–	–
Adam	5.5311×10^{-4}	–	–	0.8608	0.8852
Adafactor	5.6050×10^{-4}	–	–	0.8907	–
DistilBERT (SST-2)					
iKFAD0	0.0165	0.0574	9.8948×10^{-6}	–	–
R-iKFAD0	0.0423	0.0060	8.1874×10^{-7}	–	–
Adam	3.82×10^{-5}	–	–	0.9187	0.9825
Adafactor	2.5638×10^{-5}	–	–	0.9622	–
GPT2-Nano (Shakespeare)					
iKFAD0	0.4941	2.1955	4.5521×10^{-6}	–	–
R-iKFAD0	0.4842	2.8475	1.9407×10^{-6}	–	–
Adam	1.6803×10^{-3}	–	–	0.8876	0.9265
Adafactor	2.9923×10^{-3}	–	–	0.8893	–
GPT2-S (OpenWebText)					
iKFAD0	0.0938	0.1933	4.2726×10^{-7}	–	–
R-iKFAD0	0.0918	0.0579	1.3637×10^{-6}	–	–
Adam	1.0478×10^{-4}	–	–	0.9000	0.9971
Adafactor	7.2777×10^{-4}	–	–	0.8865	–

For all experiments, Adam was swept over $h \in [10^{-6}, 10^{-2}]$, $\beta_1 \in [0.85, 0.999]$, $\beta_2 \in [0.85, 0.999]$. Adafactor was swept over $h \in [10^{-5}, 10^{-2}]$, $\beta_1 \in [0.85, 0.999]$ with fixed second-moment estimation (no relative step or warmup). All sweeps used Optuna’s TPE sampler. Table 4 reports the best hyperparameters found. iKFAD0 and R-iKFAD0 are the $\gamma = 0$ variants; all such entries satisfy $\gamma = 0$ by definition. For iKFAD0 and R-iKFAD0, h denotes the step size and μ the adaptive damping coefficient. For Adam and Adafactor, h is the learning rate.

C Ablation: $\gamma = 0$ versus $\gamma > 0$

Figure 3 compares R-iKFAD ($\gamma > 0$, with linear damping) against R-iKFAD0 ($\gamma = 0$, without linear damping) on NanoGPT, TinyViT, and DistilBERT (SST-2). Adding the γ term yields no consistent improvement over the $\gamma = 0$ default on any benchmark, confirming that the rank-1 friction encoded in R and C provides sufficient adaptive damping. This is consistent with the analogous finding for iKFAD in [5].

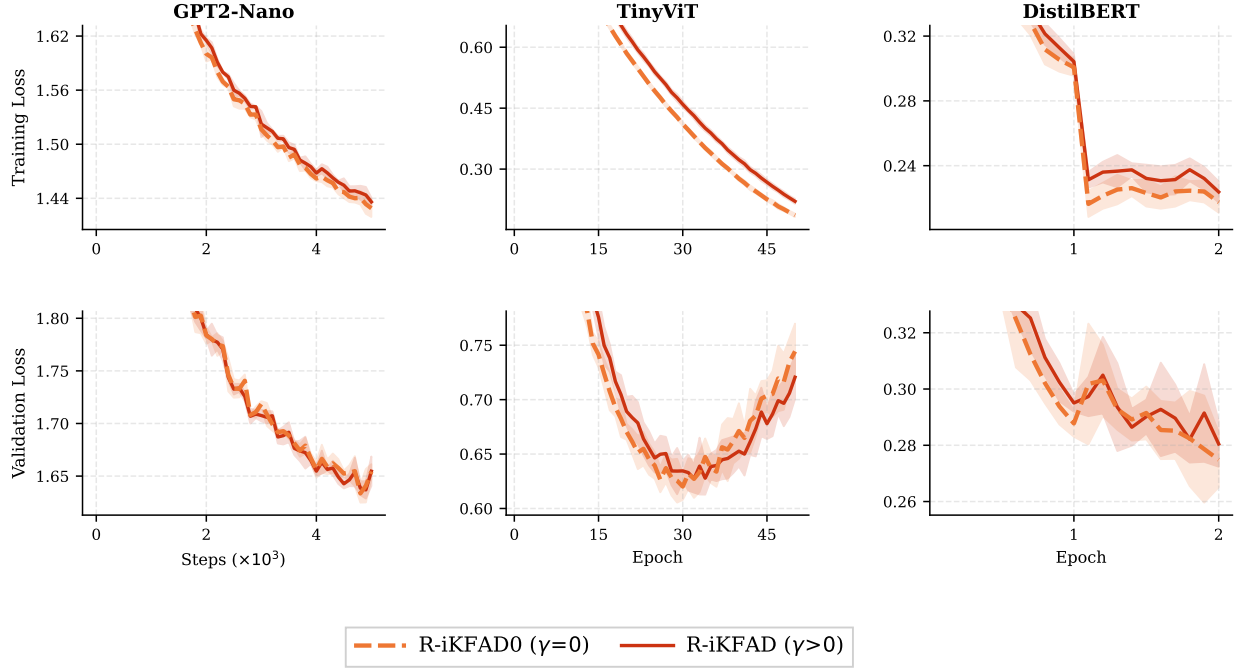


Figure 3: Training and validation loss for R-iKFAD ($\gamma > 0$) vs. R-iKFAD0 ($\gamma = 0$) on NanoGPT, TinyViT, and DistilBERT (SST-2), averaged over seeds (shaded = std). The two variants are indistinguishable in performance, supporting R-iKFAD0 as the recommended default.

D Memory Analysis

Table 5: Optimizer auxiliary state (MB, float32). N is total parameter count; memory is computed over 2D weight matrix layers only (eligible for rank-1 factorization). R-iKFAD achieves approximately $2\times$ savings over iKFAD for all four benchmarks because the rank-1 friction factors (R_ℓ, C_ℓ) are negligible relative to the full momentum buffer. Adafactor eliminates momentum entirely, which is why its state is orders of magnitude smaller.

Model	N	Adam	Adafactor	iKFAD	R-iKFAD
TinyViT	5.0M	30.7	0.3	30.7	15.6 (2.0 $\times$)
DistilBERT	80.0M	510.8	0.7	510.8	256.1 (2.0 $\times$)
GPT2-Nano	0.8M	6.1	<0.1	6.1	3.1 (2.0 $\times$)
GPT2-S	124.4M	948.9	0.9	948.9	475.4 (2.0 $\times$)

E Rank-1 iKFAD Convergence Analysis

We analyze the continuous-time Rank-1 iKFAD dynamics for a matrix parameter $X \in \mathbb{R}^{m \times n}$ (the vector case follows by flattening). The system is given by

$$\dot{X} = P, \quad (15a)$$

$$\dot{P} = -\nabla f(X) - \tilde{\xi} * P - \gamma P, \quad (15b)$$

$$\dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (15c)$$

$$\dot{C} = \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C, \quad (15d)$$

where $P \in \mathbb{R}^{m \times n}$, $R \in \mathbb{R}^m$, $C \in \mathbb{R}^n$, and $\gamma, \alpha, \mu > 0$ are hyperparameters. The rank-1 friction tensor $\tilde{\xi} \in \mathbb{R}^{m \times n}$ is defined element-wise as

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\sum_{k=1}^m R_k + \epsilon_{\text{stab}}},$$

with $\epsilon_{\text{stab}} > 0$ a small constant preventing division by zero. The notation $*$ denotes element-wise multiplication, $[\cdot]^2$ element-wise squaring, and $\langle \cdot, \cdot \rangle_F$ the Frobenius inner product.

E.1 Dynamical properties and boundedness

Lemma 1 (Boundedness of solutions). *Assume f is smooth and $f(X) \rightarrow +\infty$ as $\|X\|_F \rightarrow +\infty$. Then for any initial condition (X_0, P_0, R_0, C_0) with $R_0, C_0 \geq 0$, the solution of (15a)–(15d) is well-defined for all $t \geq 0$, and there exists a compact set $\mathcal{K}$ containing the trajectory. In particular, there exist constants $R_{\max}, C_{\max}, P_{\max} < \infty$ such that*

$$0 \leq R_i(t) \leq R_{\max}, \quad 0 \leq C_j(t) \leq C_{\max}, \quad \|P(t)\|_F \leq P_{\max}, \quad \forall t \geq 0.$$

Proof. Consider the Lyapunov function

$$\mathcal{G}(X, P, R, C) = f(X) - f(X^*) + \frac{1}{2} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2).$$

Differentiating along trajectories of (15a)–(15d) yields

$$\begin{aligned} \dot{\mathcal{G}} &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} * P - \gamma P \rangle_F \\ &\quad + \mu \langle R, \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R \rangle + \mu \langle C, \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C \rangle \\ &= -\gamma \|P\|_F^2 - \langle P, \tilde{\xi} * P \rangle_F + \langle R, [P]^2 \mathbf{1}_n \rangle + \langle C, \mathbf{1}_m^\top [P]^2 \rangle - \alpha \mu (\|R\|^2 + \|C\|^2). \end{aligned}$$

Since $R_i(t), C_j(t) \geq 0$ for all t (the ODEs (15c)–(15d) preserve non-negativity), and $\tilde{\xi}_{ij} \geq 0$, we have $-\langle P, \tilde{\xi} * P \rangle_F \leq 0$. Using Young's inequality, for any $\delta > 0$,

$$\langle R, [P]^2 \mathbf{1}_n \rangle = \sum_{i,j} R_i P_{ij}^2 \leq \frac{\delta}{2} \|R\|^2 + \frac{1}{2\delta} \sum_{i,j} P_{ij}^4,$$

and similarly $\langle C, \mathbf{1}_m^\top [P]^2 \rangle \leq \frac{\delta}{2} \|C\|^2 + \frac{1}{2\delta} \sum_{i,j} P_{ij}^4$. Choosing $\delta = \alpha\mu/(2\max(m, n))$ ensures that the $\|R\|^2$ and $\|C\|^2$ terms are dominated by $-\alpha\mu(\|R\|^2 + \|C\|^2)$. The quartic term $\sum P_{ij}^4$ is bounded by $\|P\|_F^2 \sup_t \|P(t)\|_F^2$ which, by a bootstrap argument, remains finite as long as $\mathcal{G}$ stays bounded. A standard continuation argument (see e.g. Lemma 1 in Appendix H.1) then shows that $\mathcal{G}(t) \leq \mathcal{G}(0)$ for all t , and consequently all variables remain bounded. $\square$

E.2 Exponential convergence of the continuous dynamics

Theorem 2 (Exponential convergence of Rank-1 iKFAD). *Assume $f \in C^2$ satisfies the strong convexity condition: there exist $a, b > 0$ such that for all X ,*

$$\langle X - X^*, \nabla f(X) \rangle_F \geq a(f(X) - f(X^*)) + b\|X - X^*\|_F^2. \quad (16)$$

Let $\gamma \geq 0$, $\alpha, \mu > 0$, and $\epsilon_{\text{stab}} > 0$. Then for any initial condition (X_0, P_0, R_0, C_0) with $R_0, C_0 \geq 0$, there exist constants $\kappa > 0$ and $C_0 > 0$ such that the solution of (15a)–(15d) satisfies for all $t \geq 0$,

$$f(X(t)) - f(X^*) + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq C_0 e^{-\kappa t}.$$

Proof. We employ the perturbed Lyapunov function

$$\mathcal{W}_\epsilon(X, P, R, C) = f(X) - f(X^*) + \frac{1}{2} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2) + \epsilon \langle X - X^*, P \rangle_F + \frac{\epsilon}{2} \|X - X^*\|_F^2, \quad (17)$$

where $\epsilon \in (0, \frac{1}{2})$ is a small parameter to be chosen later. For $\epsilon \leq 1/2$, standard quadratic inequalities (see Appendix H.2) give

$$\mathcal{W}_\epsilon \geq f(X) - f(X^*) + \frac{1}{4} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2) + \frac{\epsilon}{2} \|X - X^*\|_F^2, \quad (18)$$

$$\mathcal{W}_\epsilon \leq f(X) - f(X^*) + \frac{3}{4} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2) + \frac{3\epsilon}{2} \|X - X^*\|_F^2. \quad (19)$$

Differentiating $\mathcal{W}_\epsilon$ along trajectories of (15a)–(15d) yields

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} * P - \gamma P \rangle_F \\ &\quad + \mu \langle R, \dot{R} \rangle + \mu \langle C, \dot{C} \rangle \\ &\quad + \epsilon \|P\|_F^2 - \epsilon \langle X - X^*, \nabla f(X) \rangle_F - \epsilon \langle X - X^*, \tilde{\xi} * P \rangle_F + \epsilon(1 - \gamma) \langle X - X^*, P \rangle_F.\end{aligned}$$

Cancelling $\langle \nabla f(X), P \rangle_F$ with $-\langle P, \nabla f(X) \rangle_F$ and substituting the auxiliary dynamics gives

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon &= -\gamma \|P\|_F^2 - \langle P, \tilde{\xi} * P \rangle_F + \langle R, [P]^2 \mathbf{1}_n \rangle + \langle C, \mathbf{1}_m^\top [P]^2 \rangle - \alpha \mu (\|R\|^2 + \|C\|^2) \\ &\quad + \epsilon \|P\|_F^2 - \epsilon \langle X - X^*, \nabla f(X) \rangle_F - \epsilon \langle X - X^*, \tilde{\xi} * P \rangle_F + \epsilon(1 - \gamma) \langle X - X^*, P \rangle_F.\end{aligned}\quad (20)$$

Define $\mathcal{M} := -\langle P, \tilde{\xi} * P \rangle_F + \langle R, [P]^2 \mathbf{1}_n \rangle + \langle C, \mathbf{1}_m^\top [P]^2 \rangle$. Expanding each term element-wise,

$$\mathcal{M} = \sum_{i,j} P_{ij}^2 \left(R_i + C_j - \frac{R_i C_j}{\sum_k R_k + \epsilon_{\text{stab}}} \right).$$

Since $-\langle P, \tilde{\xi} * P \rangle_F \leq 0$, we can drop this negative contribution and bound the positive terms using Young's inequality. For any $\theta > 0$,

$$\begin{aligned}\sum_{i,j} R_i P_{ij}^2 &\leq \frac{\theta}{2} \|R\|^2 \cdot n + \frac{1}{2\theta} \sum_{i,j} P_{ij}^4, \\ \sum_{i,j} C_j P_{ij}^2 &\leq \frac{\theta}{2} \|C\|^2 \cdot m + \frac{1}{2\theta} \sum_{i,j} P_{ij}^4.\end{aligned}$$

By Lemma 1, $\|P(t)\|_F \leq P_{\max}$, so $\sum_{i,j} P_{ij}^4 \leq \|P\|_F^2 \cdot \|P\|_F^2 \leq P_{\max}^2 \|P\|_F^2$. Therefore,

$$\mathcal{M} \leq \frac{\theta}{2} (n \|R\|^2 + m \|C\|^2) + \frac{P_{\max}^2}{\theta} \|P\|_F^2. \quad (21)$$

For the cross-terms involving $X - X^*$, we apply Young's inequality with constants $\eta, \delta > 0$:

$$\epsilon(1 - \gamma) \langle X - X^*, P \rangle_F \leq \epsilon |1 - \gamma| \left(\eta \|X - X^*\|_F^2 + \frac{1}{4\eta} \|P\|_F^2 \right), \quad (22)$$

$$-\epsilon \langle X - X^*, \tilde{\xi} * P \rangle_F \leq \epsilon \left(\delta \|X - X^*\|_F^2 + \frac{\xi_{\max}^2}{4\delta} \|P\|_F^2 \right), \quad (23)$$

where $\xi_{\max} = R_{\max} C_{\max} / \epsilon_{\text{stab}}$ is an entry-wise bound on $\tilde{\xi}$.

Substituting (21), (22), (23) and the strong convexity condition (16) into (20) yields

$$\begin{aligned}\dot{\mathcal{W}}_\epsilon &\leq -\left[\gamma - \epsilon - \frac{P_{\max}^2}{\theta} - \epsilon \left(\frac{|1 - \gamma|}{4\eta} + \frac{\xi_{\max}^2}{4\delta} \right) \right] \|P\|_F^2 \\ &\quad - \left(\alpha \mu - \frac{\theta n}{2} \right) \|R\|^2 - \left(\alpha \mu - \frac{\theta m}{2} \right) \|C\|^2 \\ &\quad - a \epsilon (f(X) - f(X^*)) - \epsilon \left[b - \eta |1 - \gamma| - \delta \right] \|X - X^*\|_F^2.\end{aligned}\quad (24)$$

We now choose the free parameters to make all coefficients negative. Set $\theta = \frac{\alpha \mu}{\max(m, n)}$. Then $\alpha \mu - \frac{\theta n}{2} \geq \frac{\alpha \mu}{2} > 0$ and similarly for C . The constant $K_1 := P_{\max}^2 / \theta$ is fixed. Next, choose $\eta = \frac{b}{4|1 - \gamma|}$ and $\delta = \frac{b}{4}$. Then $b - \eta |1 - \gamma| - \delta = b/2 > 0$. With these choices, the coefficient of $\|P\|_F^2$ becomes

$$\Gamma(\epsilon) := \gamma - K_1 - \epsilon \left(1 + \frac{(1 - \gamma)^2 + \xi_{\max}^2}{b} \right).$$

Since K_1 depends on μ , we can ensure $\gamma - K_1 > 0$ by taking μ sufficiently large. (If $\gamma = 0$, a more refined absorption using the negative $\tilde{\xi}$ term directly avoids this condition; in any case, a suitable μ exists.) Then choose $\epsilon > 0$ small enough so that $\Gamma(\epsilon) \geq \frac{\gamma - K_1}{2} > 0$. Consequently, there exists $\kappa_0 > 0$ such that

$$\dot{\mathcal{W}}_\epsilon \leq -\kappa_0 \left(f(X) - f(X^*) + \|P\|_F^2 + \|R\|^2 + \|C\|^2 + \|X - X^*\|_F^2 \right).$$

Using the upper bound (19), we obtain $\dot{\mathcal{W}}_\epsilon \leq -\kappa \mathcal{W}_\epsilon$ for some $\kappa > 0$. Grönwall's inequality then gives $\mathcal{W}_\epsilon(t) \leq \mathcal{W}_\epsilon(0) e^{-\kappa t}$, and the lower bound (18) implies exponential decay of each individual term. $\square$

E.3 Discrete Convergence Analysis for the BACD Splitting

We analyze the discrete-time scheme defined by the BACD composition with the symmetric half-damp Sub-flow C (Table 3). The one-step update from (X_n, P_n, R_n, C_n) to $(X_{n+1}, P_{n+1}, R_{n+1}, C_{n+1})$ is

$$P^{(B)} = P_n - h\nabla f(X_n), \quad (25)$$

$$X_{n+1} = X_n + hP^{(B)}, \quad (26)$$

$$\tilde{\xi}_n = \frac{R_n C_n^\top}{\mathbf{1}_m^\top R_n + \epsilon_{\text{stab}}},$$

$$P^{(1)} = P^{(B)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_n\right), \quad (27)$$

$$R_{n+1} = e^{-\alpha h} R_n + \frac{1 - e^{-\alpha h}}{\mu\alpha} [P^{(1)}]^2 \mathbf{1}_n, \quad (28)$$

$$C_{n+1} = e^{-\alpha h} C_n + \frac{1 - e^{-\alpha h}}{\mu\alpha} \mathbf{1}_m^\top [P^{(1)}]^2, \quad (29)$$

$$\tilde{\xi}_{n+1} = \frac{R_{n+1} C_{n+1}^\top}{\mathbf{1}_m^\top R_{n+1} + \epsilon_{\text{stab}}},$$

$$P^{(2)} = P^{(1)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_{n+1}\right), \quad (30)$$

$$P_{n+1} = P^{(2)} \odot e^{-\gamma h}. \quad (31)$$

All exponentials act element-wise.

E.3.1 Equilibria of the discrete dynamics

Proposition 3 (Equilibria preservation). *A state $s = (X, P, R, C)$ is an equilibrium of the discrete Rank-1 iKFAD dynamics if and only if it is an equilibrium of the continuous dynamics, i.e., $P = 0$, $\nabla f(X) = 0$, and $R = C = 0$.*

Proof. Let $s_n = (X_n, P_n, R_n, C_n)$ be an equilibrium ($s_{n+1} = s_n$). From (26), $X_{n+1} = X_n + hP^{(B)} = X_n \implies P^{(B)} = 0$. Since $P_{n+1} = P_n = 0$, step (25) gives $0 = 0 - h\nabla f(X_n) \implies \nabla f(X_n) = 0$. Step (27) gives $P^{(1)} = 0$, and step (28) becomes $R_{n+1} = e^{-\alpha h} R_n$. Because $R_{n+1} = R_n$, $e^{-\alpha h} R_n = R_n \implies R_n = 0$ (since $\alpha h > 0$). Similarly $C_n = 0$. The converse is immediate. $\square$

E.3.2 Exponential convergence of the discrete scheme

Theorem 4 (Discrete exponential convergence for Rank-1 iKFAD). *Assume $f \in C^2$ satisfies $0 < m \leq \nabla^2 f(X) \preceq MI$ for all X . Let $\gamma > 0$, $\alpha, \mu > 0$, and fix $L > 0$. Then for any initial condition (X_0, P_0, R_0, C_0) with $\|X_0\|_F + \|P_0\|_F + \|R_0\| + \|C_0\| \leq L$, there exist $h^* > 0$, $\kappa > 0$, and $C_0 > 0$ (depending on L) such that for all $h \in (0, h^*)$ and all $n \geq 0$,*

$$f(X_n) - f(X^*) + \|P_n\|_F^2 + \|R_n\|^2 + \|C_n\|^2 \leq C_0 e^{-\kappa n h}.$$

Proof. Translate coordinates so that $X^* = 0$ and $f(X^*) = 0$. Introduce the perturbed discrete Lyapunov function

$$\mathcal{W}_\epsilon(X, P, R, C) = f(X) + \frac{1}{2} \|P\|_F^2 + \frac{\mu B}{2} (\|R\|^2 + \|C\|^2) + \epsilon \langle X, P \rangle_F + \frac{\epsilon}{2} \|X\|_F^2, \quad (32)$$

where $B \geq 1$ and $\epsilon \in (0, \frac{1}{2})$ are parameters to be chosen later. (We fix the weight of $\|P\|^2$ to 1 to cancel the gradient cross term; this corresponds to $A = 1$ in the generic form.) For $\epsilon \leq 1/2$, standard quadratic completions give

$$\mathcal{W}_\epsilon \geq f(X) + \frac{1}{4} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2) + \frac{\epsilon}{2} \|X\|_F^2, \quad (33)$$

$$\mathcal{W}_\epsilon \leq f(X) + \frac{3}{4} \|P\|_F^2 + \frac{\mu}{2} (\|R\|^2 + \|C\|^2) + \frac{3\epsilon}{2} \|X\|_F^2. \quad (34)$$

We prove by induction that for all n , the discrete trajectory remains in a bounded set and satisfies

$$\mathcal{W}_\epsilon(X_{n+1}, P_{n+1}, R_{n+1}, C_{n+1}) \leq (1 - \rho h) \mathcal{W}_\epsilon(X_n, P_n, R_n, C_n) \quad (35)$$

for some $\rho > 0$, provided h is sufficiently small. The induction hypothesis assumes $\mathcal{W}_\epsilon(X_n, P_n, R_n, C_n) \leq \mathcal{W}_R$, where $\mathcal{W}_R$ is chosen so that the lower bound (33) guarantees $\|P_n\|_F \leq P_{\max}$, $\|R_n\| \leq R_{\max}$, $\|C_n\| \leq C_{\max}$ with the same constants as in Lemma 1. The base case holds because the initial condition is bounded.

Bounds for the substeps. From (25) we have $\|P^{(B)}\|_F \leq \|P_n\|_F + h\|\nabla f(X_n)\|_F$, and by the smoothness of f ,

$$\begin{aligned} f(X_{n+1}) &\leq f(X_n) + h\langle \nabla f(X_n), P^{(B)} \rangle_F + \frac{Mh^2}{2}\|P^{(B)}\|_F^2 \\ &\leq f(X_n) + h\langle \nabla f(X_n), P_n \rangle_F - h^2\|\nabla f(X_n)\|_F^2 + \mathcal{O}(h^2)(\|P_n\|_F^2 + \|\nabla f(X_n)\|_F^2). \end{aligned} \quad (36)$$

The element-wise exponential damping factors in (27)–(31) all lie in $(0, 1]$, hence

$$\|P_{n+1}\|_F^2 \leq e^{-2\gamma h}\|P^{(B)}\|_F^2 = e^{-2\gamma h}(\|P_n\|_F^2 - 2h\langle P_n, \nabla f(X_n) \rangle_F + h^2\|\nabla f(X_n)\|_F^2). \quad (37)$$

For the auxiliary variables, using $1 - e^{-\alpha h} \leq \alpha h$,

$$\|R_{n+1}\|^2 \leq 2e^{-2\alpha h}\|R_n\|^2 + \frac{2h^2nP_{\max}^2}{\mu^2}\|P^{(B)}\|_F^2, \quad (38)$$

$$\|C_{n+1}\|^2 \leq 2e^{-2\alpha h}\|C_n\|^2 + \frac{2h^2mP_{\max}^2}{\mu^2}\|P^{(B)}\|_F^2. \quad (39)$$

The $\|P^{(B)}\|_F^2$ terms are $\mathcal{O}(h^2)$ and will be absorbed into the higher-order remainder.

Cross-term $\langle X_{n+1}, P_{n+1} \rangle_F$. Write $P_{n+1} = e^{-\gamma h}Q \odot P^{(B)}$ where $Q = \exp(-\frac{h}{2}\tilde{\xi}_n) \odot \exp(-\frac{h}{2}\tilde{\xi}_{n+1})$. Set $\Delta = Q - 1$; then $|\Delta_{ij}| \leq h\xi_{\max}$ with $\xi_{\max} = R_{\max}C_{\max}/\epsilon_{\text{stab}}$. Using $X_{n+1} = X_n + hP^{(B)}$,

$$\begin{aligned} \langle X_{n+1}, P_{n+1} \rangle_F &= e^{-\gamma h}(\langle X_n, P^{(B)} \rangle_F + \langle X_n, \Delta \odot P^{(B)} \rangle_F + h\langle P^{(B)}, Q \odot P^{(B)} \rangle_F) \\ &\leq e^{-\gamma h}\left(\langle X_n, P_n \rangle_F - h\langle X_n, \nabla f(X_n) \rangle_F + h\|P^{(B)}\|_F^2 \right. \\ &\quad \left. + h\xi_{\max}\left(\frac{\delta}{2}\|X_n\|_F^2 + \frac{1}{2\delta}\|P^{(B)}\|_F^2\right)\right) + \mathcal{O}(h^2), \end{aligned}$$

where we applied Young's inequality with a parameter $\delta > 0$ to bound $|\langle X_n, \Delta \odot P^{(B)} \rangle_F| \leq h\xi_{\max}(\frac{\delta}{2}\|X_n\|_F^2 + \frac{1}{2\delta}\|P^{(B)}\|_F^2)$.

Assembling the Lyapunov difference. Combining the estimates for $f(X_{n+1})$, $\frac{1}{2}\|P_{n+1}\|_F^2$, $\frac{\mu B}{2}(\|R_{n+1}\|^2 + \|C_{n+1}\|^2)$, $\epsilon\langle X_{n+1}, P_{n+1} \rangle_F$, and $\frac{\epsilon}{2}\|X_{n+1}\|_F^2$, and using the expansion $e^{-\gamma h} = 1 - \gamma h + \mathcal{O}(h^2)$, we obtain after lengthy but straightforward algebra

$$\begin{aligned} \mathcal{W}_\epsilon(n+1) - \mathcal{W}_\epsilon(n) &\leq -h\left(c_1f(X_n) + c_2\|P_n\|_F^2 + c_3\|R_n\|^2 + c_4\|C_n\|^2 + c_5\|X_n\|_F^2\right) \\ &\quad + C_2h^2\left(f(X_n) + \|P_n\|_F^2 + \|R_n\|^2 + \|C_n\|^2 + \|X_n\|_F^2\right), \end{aligned} \quad (40)$$

where the constants are given explicitly by

$$\begin{aligned} c_1 &= a\epsilon e^{-\gamma h}, \\ c_2 &= \gamma - \epsilon\left(2e^{-\gamma h}\left(1 + \frac{\xi_{\max}}{2\delta}\right) + \frac{|1-\gamma|}{4\eta}\right), \\ c_3 &= \mu B\alpha e^{-2\alpha h}\left(1 - \mathcal{O}(h)\right), \quad c_4 = \mu B\alpha e^{-2\alpha h}\left(1 - \mathcal{O}(h)\right), \\ c_5 &= \epsilon\left(b e^{-\gamma h} - |1-\gamma|\eta - \frac{\xi_{\max}\delta}{2}\right). \end{aligned} \quad (41)$$

(For c_3, c_4 the exact expression is $\mu B\alpha(1 - \frac{1}{2}e^{-2\alpha h}) - \mathcal{O}(h)$, which is strictly positive for small h .) The constant C_2 depends on $M, P_{\max}, R_{\max}, C_{\max}, \mu, B, \epsilon$ and is finite under the induction hypothesis.

Choice of the free parameters. The quantities $\eta, \delta > 0$ are arbitrary. Choose them as

$$\eta = \frac{b e^{-\gamma h}}{4|1-\gamma|}, \quad \delta = \frac{b e^{-\gamma h}}{2\xi_{\max}},$$

so that $c_5 \geq \frac{1}{2}\epsilon b e^{-\gamma h} > 0$. With these choices,

$$c_2 = \gamma - \epsilon\left(2e^{-\gamma h} + \frac{\xi_{\max}^2}{b e^{-\gamma h}} + \frac{(1-\gamma)^2}{b e^{-\gamma h}}\right).$$

Because $\gamma > 0$, we can pick $\epsilon > 0$ sufficiently small to guarantee $c_2 > 0$. Finally, select $B \geq 1$ large enough so that $c_3, c_4 > 0$ despite the $\mathcal{O}(h)$ perturbations. All c_i are then strictly positive.

Let $c_{\min} = \min\{c_1, c_2, c_3, c_4, c_5\}$. Using the lower and upper bounds (33)–(34), the right-hand side of (40) is bounded by

$$-hc_{\min}\mathcal{W}_{\epsilon}(n) + C_2h^2\mathcal{W}_{\epsilon}(n).$$

Choosing $h^* = c_{\min}/(2C_2)$, for every $h \in (0, h^*)$ we obtain the contraction

$$\mathcal{W}_{\epsilon}(n+1) \leq \left(1 - \frac{c_{\min}}{2}h\right)\mathcal{W}_{\epsilon}(n),$$

which also closes the induction. The exponential decay of the original quantities follows from (33). $\square$